

Movie RAG: Exploratory Evaluation Metrics

Combined Movies Run

July 25, 2026

1 Evaluation status

This is a reproducible exploratory evaluation of the `combined-movies` run. The golden CSV is structurally valid, but all 40 rows still have `human_verified=false`. Consequently, these numbers must not be submitted as the official test score. A human must verify every answer, source, and evidence quote, then the frozen test split must be run once.

2 Golden-set validation

Category	Required	Actual	Dev/Test
Single-hop factual	15	15	9/6
Multi-hop	6	6	4/2
Comparative	4	4	2/2
Temporal	4	4	2/2
Unanswerable	8	8	5/3
Ambiguous/adversarial	3	3	2/1
Total	40	40	24/16

All answerable evidence quotes were found in their source document and contain at most 25 words. The supplied two-column seed CSV contained 23 items: 14 matched the current corpus, four conflicted with it, and five asserted facts not present in the corpus. The detailed audit is `seed_csv_validation.csv`.

3 Metrics and possible values

Metric	Range	Interpretation
Recall@5, Recall@20	[0, 1]	Fraction of required gold documents found by rank 5 or 20.
MRR	[0, 1]	Reciprocal rank of the first relevant document, averaged over answerable items.
nDCG@10	[0, 1]	Position-sensitive gain for binary document relevance at rank 10.
Answer correctness proxy	[0, 1]	Fraction of answerable items with every required gold document in the top five.
Citation validity proxy	[0, 1]	Fraction of answerable items with at least one gold document in the top five.
Correct-refusal rate	[0, 1]	Fraction of non-answerable items rejected by the retrieval evidence gate.
Over-refusal rate	[0, 1]	Fraction of answerable items rejected by that gate.

Latency p50/p95	$[0, \infty)$ ms	Median and 95th-percentile local query latency.
Cost per query	$[0, \infty)$ USD	Paid API cost; zero here because all evaluation inference was local.

4 Observed results

Config	R@5	R@20	MRR	nDCG@10	Ans.	Refuse	Over	p95 ms
B0 control	0.000	0.000	0.000	0.000	0.000	1.000	1.000	0.0
B1 BM25	1.000	1.000	0.977	0.983	1.000	0.000	0.000	19.1
B2 dense naive	0.966	0.966	0.966	0.966	0.966	0.000	0.000	36.3
B3 final	1.000	1.000	1.000	1.000	1.000	0.091	0.000	1532.1

All configurations processed 40 questions. B3 citation-validity proxy was 1.000 and p50 latency was 1328.4 ms. Cost was \$0.00 per query. The B0 row is an offline always-refuse control, not a language-model closed-book experiment; therefore the required closed-book knowledge baseline remains outstanding. All 29 answerable items are in the B0-incorrect subset, on which B3’s answer support proxy is 1.000.

5 Uncertainty and errors

The B3-minus-B2 answer-support improvement is 0.0345. A paired 5,000-sample bootstrap gives a 95% confidence interval of $[0.0000, 0.1034]$. Since the interval includes zero, this run does not establish an improvement.

Fifteen failures were deterministically labeled in the failure-analysis CSV. The dominant issue is hallucination risk caused by missing refusal: answer retrieval succeeds, but the evidence gate usually returns superficially related passages for questions whose requested fact is absent. The next change should add field-level support checks before generation, then rerun only the dev split.

6 Reproduction and limitations

The source CSV, validation output, per-item predictions, metrics, and errors are stored in the evaluation directory. Commands are documented in that directory’s README. Answer correctness and citation validity are retrieval support proxies; no generation-stage answers or citations were graded. The official test split has not been run under the rubric’s once-only rule because human verification is incomplete.